Evaluating Local LLM Pipelines for Ontology Generation

Bench4KE: Ollama Support, Documentation Completeness, and Prompt Sensitivity

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Abstract

This project evaluates three ontology-generation pipelines—Ontogenia, Domain-OntoGen, and NeOn-GPT—using one fully local Ollama model and an evidence-preserving execution contract. The experiment covers 17 ontology-engineering scenarios, 74 source-ordered competency questions (CQs), and three prompt variants: the recovered baseline P0, serialization-only P1, and syntax-verification P2. Across 153 admitted tasks, 101 final ontologies were parseable. Final parse success was 14/51 for Ontogenia, 50/51 for Domain-OntoGen, and 37/51 for NeOn-GPT. Repair-aware paired analysis found no statistically significant parse-success difference among P0, P1, and P2 for any method. This result does not imply prompt invariance: descriptive term-Jaccard summaries show substantial ontology-content drift from P0 where comparisons are available. Documentation completeness and parse robustness also varied materially by method. The contribution is both empirical and methodological: it supplies the mandatory open-model execution path and two new evaluation components while keeping model behavior, transformations, and evidence governance separately auditable.

1 Introduction

Large language models can turn scenarios and competency questions into candidate ontologies, but a defensible benchmark must evaluate more than whether a final file exists. Prompts, intermediate outputs, parsing transformations, model telemetry, and cache identity must be preserved so that errors and later analytical decisions remain auditable. Bench4KE provides a starting point for standardized knowledge-engineering evaluation [1], while Ontogenia and related work establish three generation procedures used in this project [5, 7].

The course brief requires mandatory Ollama support, at least two additional evaluation components, and testing on the entire executable normalized dataset with the three methods already connected to Bench4KE. This submission implements all three requirements. It compares the methods under one exact local model and adds documentation completeness and prompt sensitivity as deterministic, reproducible analyses.

2 Related Work

Recent work has investigated the use of large language models for ontology engineering and ontology drafting from natural-language requirements. Ontogenia introduced a metacognitive prompting strategy for ontology generation with large language models, using competency questions as part of the ontology-engineering input [5]. Subsequent work studied LLM-based OWL ontology drafting from user stories and competency questions and evaluated prompting techniques such as Memoryless CQbyCQ and Ontogenia [7]. A separate domain-specific study evaluated DeepSeek and o1-preview on 95 curated CQs from six domains; its public companion resource supplies the prompt recovered for the Domain-OntoGen baseline in this project [6]. NeOn-GPT represents a related LLM-powered ontology-learning pipeline grounded in the NeOn methodology and designed to translate natural-language domain descriptions into Turtle ontologies [2]. Bench4KE provides an extensible benchmark-oriented setting for knowledge-engineering automation, with its first release focused on automatic competency-question generation and with other KE tasks left as possible extensions [1]. The present work uses three generation methods available in the project baseline: Ontogenia, Domain-OntoGen, and NeOn-GPT. Its goal is to extend their evaluation rather than to propose a new ontology-generation method.

Ontology documentation is a separate concern from syntactic validity and axiomatic structure. An ontology may be parseable while still being difficult for humans to inspect, maintain, or reuse because its classes and properties lack labels, comments, definitions, or ontology-level metadata. Tools such as WIDOCO treat documentation as an explicit ontology-engineering activity by detecting missing vocabulary metadata and generating human-readable documentation and diagrams [4]. However, documentation support tools do not directly provide a benchmark for measuring the documentation produced by LLM-based ontology generators. The C2 component of this project therefore measures the coverage of labels, comments or definitions, descriptive metadata, and nontrivial documentation literals in generated ontologies. These metrics measure documentation presence and coverage; they do not establish the semantic correctness of the annotations or of the ontology itself.

Prompt robustness has also become an important concern in LLM evaluation. PromptBench studies the robustness of large language models to adversarial prompt perturbations, including character-level, word-level, sentence-level, and semantic-level changes [8]. Such sensitivity is particularly relevant to ontology generation because a response must satisfy both domain-modeling requirements and a strict serialization grammar. Nevertheless, the C3 experiment in this project is narrower than general adversarial or semantic prompt-robustness evaluation. P1 and P2 are append-only instruction suffixes applied to the recovered P0 prompts. Scenarios, competency questions, examples, method procedures, call structures, parsing policies, and generation controls remain fixed. C3 consequently evaluates instruction-level suffix sensitivity through paired parse-success outcomes and ontology-term Jaccard overlap; it does not evaluate unrestricted semantic rephrasing.

The project specification additionally identifies local open-model execution as a missing benchmark capability. Local execution can reduce dependence on paid proprietary APIs, but locality alone does not guarantee reproducibility. A reproducible evaluation must also preserve the runtime version, exact model tag and digest, generation controls, prompts, responses, telemetry, parse stages, repair behavior, and cache identity. The Ollama integration in this project adds this evidence-preserving execution layer to the three generation methods. The empirical conclusions remain limited to one quantized Qwen model, one seed, and one fixed context and output budget, and therefore should not be generalized to all open models.

In summary, the project brings together three complementary concerns: LLM-based ontology generation, ontology documentation, and prompt robustness. Its contribution is an extension of the project benchmark with local Ollama execution, deterministic documentation-completeness metrics, and a narrowly controlled prompt-sensitivity analysis. It does not claim to solve general ontology-quality evaluation or broad prompt robustness.

3 Objectives and Contributions

The project delivers three benchmark extensions:

1. **C1—native local Ollama execution and evidence preservation.** The provider records exact model identity, fixed generation controls, requests, responses, per-step telemetry, parsing stages, errors, and versioned cache keys.
2. **C2—documentation completeness.** A deterministic evaluation measures labels, comments or definitions, ontology metadata, documentation length, and nontrivial documentation for parseable generated ontologies.
3. **C3—prompt sensitivity.** Frozen P0/P1/P2 variants are compared through paired parse-success tests and ontology-term Jaccard overlap under a prespecified statistical policy.

All model generation used a local open model. No paid API was required, and no model call was made during final statistical analysis or report preparation.

4 Dataset Construction

The normalized full-generation dataset contains 17 scenarios and 74 executable CQs. The source workbook contains 112 nonempty CQ rows. Thirty-eight rows are retained in audit records but excluded from executable generation because they lack reliable StoryID or scenario linkage. The pipeline does not infer or fuzzily reconstruct missing links.

Scenarios, user stories, and CQs were recovered from the fixed workbook through explicit source relationships. CQ wording and source order are preserved. Every managed data file was validated with its expected parser and encoding, and the normalized JSONL was explicitly audited for UTF-8 integrity. The frozen dataset manifest is:

`e06831a155503aa5c2faa8312b7bd78eb6778b124f31dbfb1617bc63c6664caf`

This conservative construction favors traceability over maximizing row count. It also makes missingness visible instead of silently inventing relationships.

5 Compared Generation Methods

5.1 Ontogenia

Ontogenia generates CQ-level ontology fragments and combines them into a task-level ontology. Its multi-call structure makes previous-output propagation and call order part of the executable method definition.

5.2 Domain-OntoGen

Domain-OntoGen processes CQs independently through the approved domain-oriented procedure recovered from the public companion resource for the domain-specific ontology-generation study [6], then assembles the resulting ontology. In the experiment it is the most consistently parseable method, although high parseability does not imply rich documentation or conceptual correctness.

5.3 NeOn-GPT

NeOn-GPT follows the frozen staged NeOn procedure. After a parse failure it may invoke deterministic syntax-repair prompts. Those repair calls are recorded as variant-agnostic post-processing rather than hidden generation steps.

5.4 Prompt variants

P0 is the authoritative recovered baseline. P1 appends an instruction to return only the requested serialization. P2 adds silent checks for prefix declarations, complete RDF statements, well-formed lists and blank nodes, and consistent class and property naming. Scenarios, CQs, examples, procedures, call order, parsing, normalization, repair logic, and generation controls are otherwise unchanged.

6 Local Ollama Infrastructure

All admitted A1 and A2 evidence uses the following frozen configuration:

- Ollama version 0.32.0;
- model `qwen3:30b-a3b-instruct-2507-q4_K_M`;
- digest `19e422b0231392335cfc49cfd172de7034bb1aeabb08aa307cce745c60b272fe`;
- temperature 0 and seed 42;
- `num_ctx=32768`, `num_predict=8192`, `keep_alive=30m`, and non-streaming responses.

The preserved A1 and A2 preflight records provide hardware context for the runtime measurements: they identify one NVIDIA GeForce RTX 5090 with 32,607 MiB total GPU memory and 100,555,055,104 bytes of total system RAM. Available memory varied between checks. The records do not preserve the CPU model, exact operating-system release, or GPU-placement telemetry for the admitted calls at `num_ctx=32768`; no claims are therefore made about those properties or about hardware-independent performance.

The provider verifies the Ollama version and exact model tag/digest before a run. For every executed internal call, the evidence contract preserves the executed prompt, serialized request and response, raw text, normalized text, final text, parse status, error record, prompt and completion counts, timing, completion reason, HTTP attempt, and previous-output input when applicable. Resume behavior uses cache identity and holds failed tasks unless an explicitly authorized retry mode is selected.

Across admitted A1 and A2 evidence, 576 internal calls and HTTP attempts were recorded, with zero retries. Telemetry totals were 1,467,185 prompt tokens, 1,167,385 completion tokens, and 4,462.483 summed task seconds. These are local measurements, not monetary API costs.

7 Output Contract and Evidence Admission

In this report, a *result envelope* is the persisted per-task bundle of status, outputs, parse records, telemetry, and cache-identity metadata. A *sidecar* is a separate audit record that preserves the original envelope while recording identity differences and an analytical decision. *Evidence admission* means that immutable artifacts satisfy an explicit rule for downstream analysis; it does not convert a historical cache-identity failure into a successful execution contract.

Execution success and analytical usability are distinct. Stage A1 failed its original schema-v1 cache-identity contract because numerically equivalent generation parameters had different serialized representations. Its immutable evidence was subsequently admitted through a schema-v2 semantic-equivalence audit.

Stage A2 retains three explicitly distinct historical layers:

- historical Stage A2 strict schema-v2 identity: **FAIL**;
- Phase 12 strict sidecar admission: **PARTIALLY ADMITTED**;
- Phase 12B repair-aware sidecar admission: **ADMITTED**.

The A2 defect was a success-path metadata overwrite: preserved executed P1/P2 prompts were correct, but 94 successful envelopes stored the corresponding method P0 `prompt_hash`. The repair-aware ruling separately classifies 53 exact frozen NeOn syntax-repair calls as variant-agnostic post-processing. Historical envelopes and raw outputs were never rewritten. Thus downstream admission does not retroactively turn the strict execution contract into a pass.

8 Evaluation Metrics

8.1 Parse and structural metrics

Raw, normalized, and final Turtle are parsed separately. Triple, class, object property, and datatype property counts are calculated only for parseable final ontologies. Failed parses remain unavailable observations rather than being encoded as empty ontologies.

8.2 C2 documentation completeness

Labels are identified through `rdfs:label` or `skos:prefLabel`. Documentation is identified through `rdfs:comment`, `skos:definition`, `dcterms:description`, or `schema:description`. A nontrivial documentation literal contains at least 20 Unicode characters and three whitespace-delimited words. Entity-level C2 coverage is conditional on parseable output, while metric availability is always reported over all 153 tasks.

8.3 C3 prompt sensitivity

C3 uses paired final-parse outcomes. Cochran's Q is the omnibus test. Pairwise McNemar tests with Holm correction are interpreted only under the frozen gatekeeping policy. Content sensitivity compares P0 with P1 and P2 through ontology-term Jaccard overlap. Paired Pratt Wilcoxon tests compare whether P1 and P2 have different drift magnitudes from P0 where enough complete cases exist. The latter test does not test whether either variant differs from P0.

9 Experimental Design

Stage A1 contains 51 admitted P0 tasks: 17 dataset items multiplied by three methods. Stage A2 contains 102 repair-aware-admitted tasks: the same 17 items and three methods for P1 and P2. Together, the analysis panel contains 153 tasks.

The model, digest, seed, temperature, context budget, output budget, prompt/CQ order, method procedures, normalization, parsing, and repair policy were frozen before execution. A1 P0 serves as the admitted baseline. A2 changes only the approved instruction-level suffix. This design supports within-item comparisons while preserving the original P0 evidence.

10 Results

10.1 Parse success

Table 1: Final parse success by method and prompt variant.

Method	P0	P1	P2
Ontogenia	4/17	5/17	5/17
Domain-OntoGen	16/17	17/17	17/17
NeOn-GPT	11/17	12/17	14/17

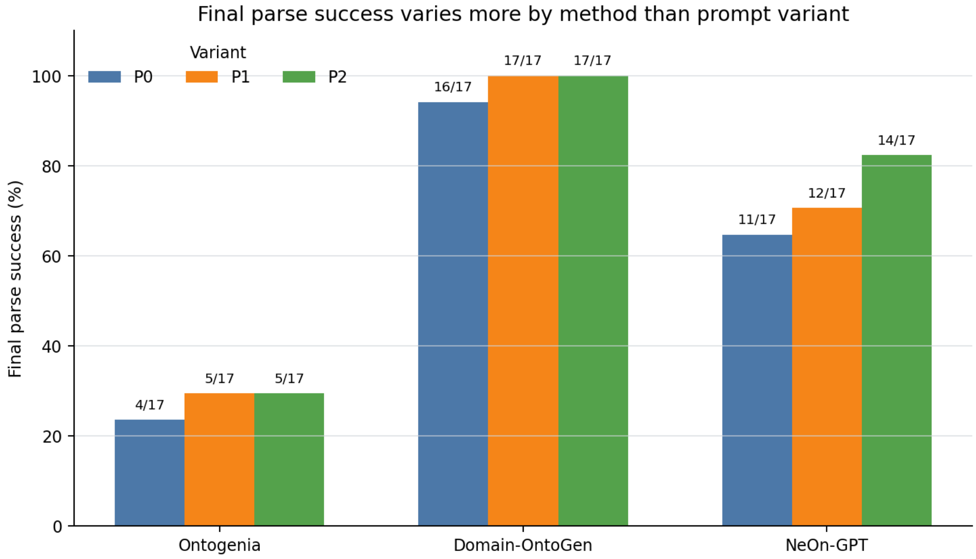

Figure 1: Final parse success by method and prompt variant.

Across all variants, final parse success was 14/51 for Ontogenia, 50/51 for Domain-OntoGen, and 37/51 for NeOn-GPT. Method-level differences are therefore much larger than the observed within-method variant differences. Normalization and syntax repair improve final parseability, particularly for Ontogenia and NeOn-GPT.

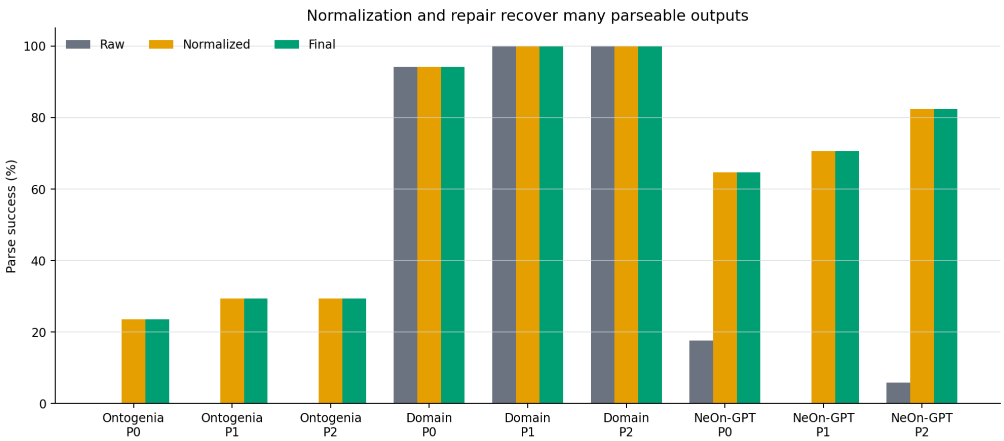

Figure 2: Raw, normalized, and final parse success.

10.2 Repairs and length terminations

The combined experiment records 81 repair calls and 41 `done_reason=length` terminations. Repairs are concentrated in NeOn-GPT. Length stops also occur in Ontogenia P1/P2 and Domain-OntoGen P0. These events are retained as evidence rather than silently excluded.

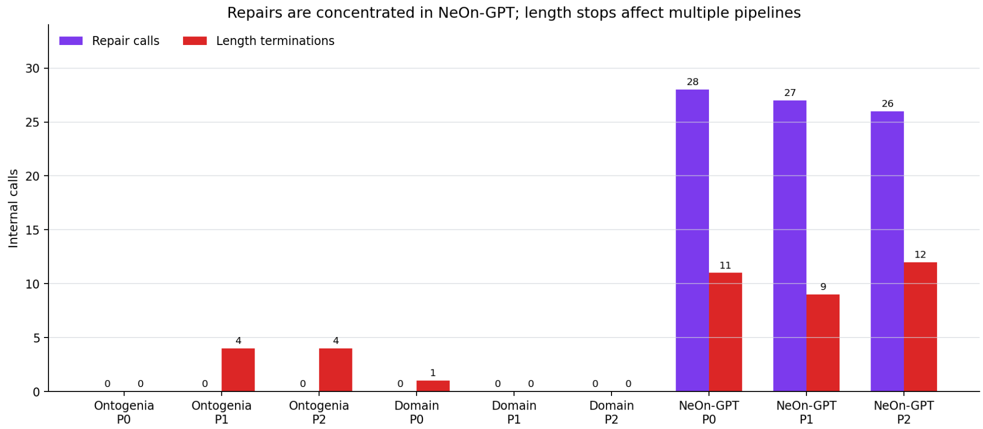

Figure 3: Repair calls and length terminations by method and variant.

10.3 C2 documentation completeness

C2 is available for 101/153 tasks (66.0%); the 52 unparseable outputs remain explicitly unavailable. Across parseable ontologies, class-label coverage is 77.3% and class-documentation coverage is 43.9%. Object-property label and documentation coverage are 85.5% and 49.1%, while datatype-property label and documentation coverage are 68.7% and 60.7%. Only one parseable task contains the required ontology-metadata declaration and descriptive statement.

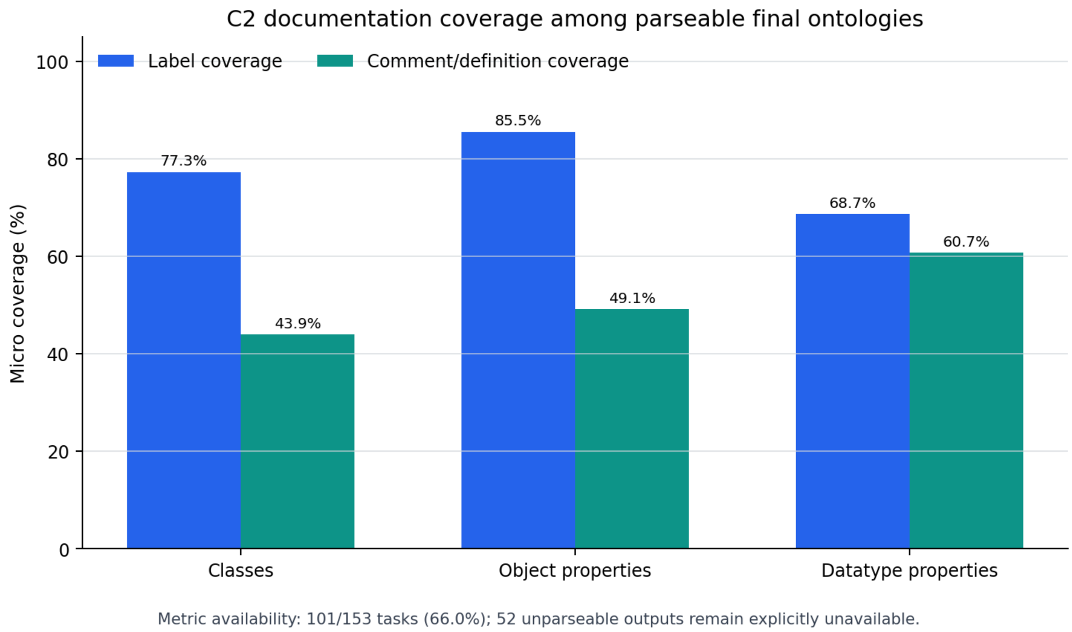

Figure 4: Documentation coverage conditional on parseable final ontologies.

These values are descriptive, not semantic-quality scores. Domain-OntoGen has very high label coverage but almost no comments or definitions. NeOn-GPT and parseable Ontogenia outputs often contain richer documentation despite lower overall parse availability.

10.4 C3 parse-success inference

Table 2: C3 paired statistical summary.

Method	Triplets	Q p	Term pairs	Wilcoxon p	Interpretation
Ontogenia	17	0.9048	0	*NA*	No significant parse difference; drift comparison not estimable.
Domain-OntoGen	17	0.3679	16	0.3380	No significant parse difference; P1/P2 drift magnitude not distinguishable.
NeOn-GPT	17	0.5292	6	0.1957	No significant parse difference; small paired drift sample.

Cochran Q p-values are 0.9048 for Ontogenia, 0.3679 for Domain-OntoGen, and 0.5292 for NeOn-GPT. Every Holm-adjusted McNemar p-value is 1.0. Under the frozen policy, none of the methods shows a statistically significant P0/P1/P2 final parse-success difference.

10.5 C3 descriptive term drift

Table 3: Term-Jaccard overlap with P0. Lower values indicate more drift.

Method	Comparison	Eligible n	Median	Min	Max
Ontogenia	$J(P0, P1)$	0	*NA*	*NA*	*NA*
Ontogenia	$J(P0, P2)$	1	0.2667	0.2667	0.2667
Domain-OntoGen	$J(P0, P1)$	16	0.5682	0.3077	1.0000
Domain-OntoGen	$J(P0, P2)$	16	0.6500	0.3333	1.0000
NeOn-GPT	$J(P0, P1)$	8	0.1159	0.0000	0.4000
NeOn-GPT	$J(P0, P2)$	8	0.0000	0.0000	0.2222

The Domain-OntoGen and especially NeOn-GPT summaries show substantial descriptive ontology-content drift from P0. Ontogenia has no P0/P1 comparison and only one P0/P2 value, so no distributional conclusion is justified for that method. Domain-OntoGen has 16 complete P1/P2 drift pairs (Wilcoxon p=0.3380), NeOn-GPT has six (p=0.1957), and Ontogenia has none. The NeOn-GPT test uses the frozen asymptotic Pratt approximation and has low power.

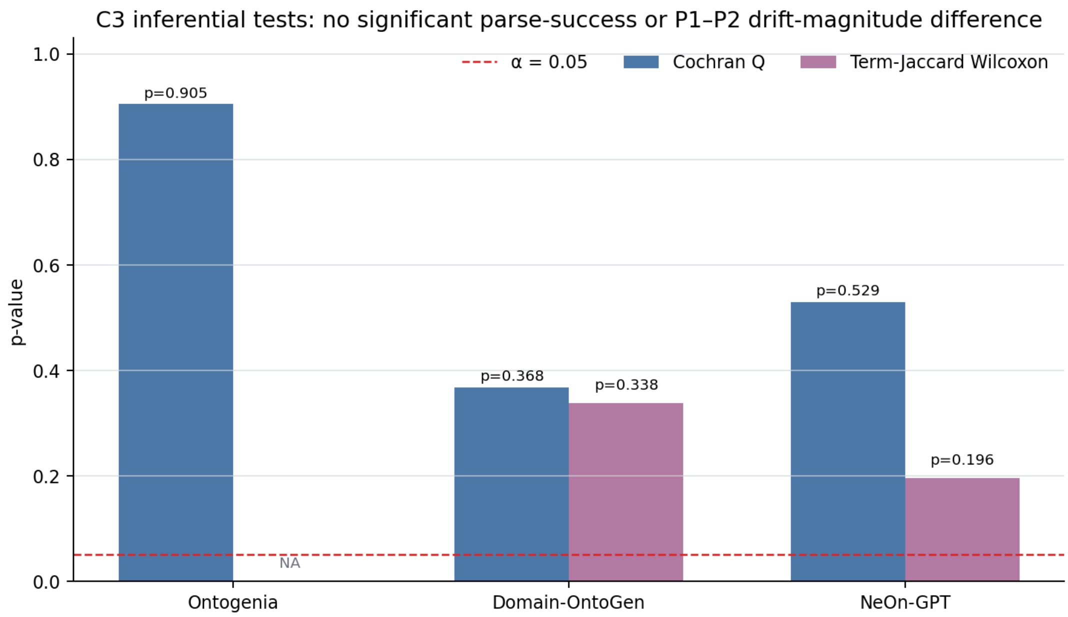

Figure 5: Prompt-sensitivity summary: parse inference and descriptive term drift.

The correct conclusion is deliberately scoped: prompt variants did not show statistically significant parse-success differences under the repair-aware admitted analysis, but term overlap still indicates substantial descriptive content drift. The absence of parse significance is not evidence of prompt invariance or superiority of P1/P2.

11 Discussion

Domain-OntoGen is the most consistently parseable method across all variants, but its generated ontologies provide little class or property documentation. NeOn-GPT has intermediate parse robustness and richer documentation, with additional repair cost and truncation exposure. Ontogenia has the lowest task-level parse robustness because many CQ fragments contain invalid Turtle. The remaining Ontogenia issue is monitored as model-output quality risk rather than a demonstrated deterministic prompt-assembly defect.

P1 and P2 show small descriptive increases in parse success for some cells, especially NeOn-GPT P2, but the paired tests do not distinguish these changes from chance under the frozen policy. Independently, low term-Jaccard overlap shows that content can change substantially even when parse-success differences are not statistically significant. Prompt sensitivity therefore cannot be reduced to syntax alone.

The evidence history also demonstrates why a benchmark needs explicit data and execution contracts. A valid model response can coexist with an invalid cache identity; a repaired ontology can be parseable without being a raw parse success; and an analytically admitted artifact can retain a historical execution failure. Keeping these concepts separate prevents silent overclaiming.

12 Failure Analysis

Three failure classes should not be conflated:

1. **Model-output quality:** malformed Turtle, incompatible CQ fragments, or truncated responses.

2. **Post-processing and repair:** normalization and frozen syntax repair may recover parseability but must remain visible in telemetry.
3. **Evidence identity:** A1 numeric canonicalization and A2 prompt-hash substitution affected cache/audit contracts even where requests and responses were preserved.

The evidence-identity failures are handled under the rules in Section 7; neither analytical admission nor the writer correction altered preserved requests, responses, or outputs.

13 Limitations

- Only one local quantized model, one seed, and one context/output budget were evaluated.
- The dataset contains 17 scenarios; some paired term denominators are small or zero.
- Parse success does not establish conceptual correctness, CQ coverage, logical consistency, or practical ontology usefulness.
- C2 is conditional on syntactically parseable output and does not judge semantic quality of documentation.
- Normalization and repair affect observed final parseability.
- Repair-aware admission is an analysis-governance decision and does not make the historical strict Stage A2 contract pass.
- Public source ontologies may create contamination risk; the current work does not claim to eliminate that risk.

14 Reproducibility and Submission Artifacts

The full research workspace retains the frozen dataset, recovered prompts, model digest, generation controls, task plans, requests, raw responses, parse stages, telemetry, cache keys, and admission sidecars. The clean submission deliberately excludes large raw-output trees while retaining the implementation and compact final aggregates required to inspect the experiment.

Key submitted artifacts include:

- `restapi/`: provider, adapter, API, metrics, and artifact logic;
- `scripts/`: deterministic preparation, execution, and analysis tools;
- `datasets/ontology_generation/`: frozen normalized data and prompts;
- `config/`: frozen experiment and statistical policies;
- `results/final_experiment_manifest.json`: integrated hashes and counts;
- `results/final_c2_c3_summary.csv`: final method-level results;
- `report/figures/figure_manifest.json`: figure provenance;

Re-running model generation requires the exact Ollama model and constitutes a new execution. Deterministic code and results can be inspected without a live model or proprietary provider. The repository implementation is based on and extends the Bench4KE project [3].

15 Conclusion

The project satisfies the course scope by implementing mandatory local Ollama support and two additional evaluation components, then applying the three required ontology-generation methods to the complete frozen 17-item dataset. The infrastructure produced auditable local evidence and a deterministic final analysis.

Domain-OntoGen achieved the highest parse robustness, NeOn-GPT combined intermediate parseability with richer documentation and higher repair exposure, and Ontogenia remained the least robust at task-level Turtle parsing. P0/P1/P2 did not differ significantly in paired parse-success tests. However, descriptive term-Jaccard overlap showed substantial content drift, especially for NeOn-GPT, so non-significance must not be read as prompt invariance.

The strongest contribution is a reproducible evaluation framework that keeps model behavior, transformation behavior, and evidence-governance decisions separate and reviewable. This supports more credible future comparisons across models, prompts, and ontology-engineering tasks.

A Detailed Parse Summary

Table 4: Raw, normalized, and final parse counts.

Method	Variant	Tasks	Raw	Normalized	Final
Ontogenia	P0	17	0	4	4
Ontogenia	P1	17	0	5	5
Ontogenia	P2	17	0	5	5
Domain-OntoGen	P0	17	16	16	16
Domain-OntoGen	P1	17	17	17	17
Domain-OntoGen	P2	17	17	17	17
NeOn-GPT	P0	17	3	11	11
NeOn-GPT	P1	17	0	12	12
NeOn-GPT	P2	17	1	14	14

B Generation Cost and Runtime Summary

Table 5: Internal-call telemetry by method and variant.

Method	Variant	Calls	Prompt tokens	Completion tokens	Total tokens	Seconds
Ontogenia	P0	74	160,584	110,780	271,364	532.761
Ontogenia	P1	74	163,322	122,812	286,134	594.151
Ontogenia	P2	74	166,282	136,927	303,209	651.114
Domain-OntoGen	P0	74	169,142	70,871	240,013	337.428
Domain-OntoGen	P1	74	171,954	49,234	221,188	249.039
Domain-OntoGen	P2	74	174,914	55,740	230,654	268.731
NeOn-GPT	P0	45	165,240	226,560	391,800	540.968
NeOn-GPT	P1	44	149,032	196,769	345,801	528.101
NeOn-GPT	P2	43	146,715	197,692	344,407	760.189
Total	**All**	**576**	**1,467,185**	**1,167,385**	**2,634,570**	**4,462.483**

C Evidence Status Summary

Table 6: Historical status and downstream disposition.

Item	Historical or strict status	Downstream disposition
Stage A1	Schema-v1 identity: **FAIL**	Immutable evidence admitted through schema-v2 semantic-equivalence audit.
Stage A2	Strict schema-v2 identity: **FAIL**	Historical failure preserved.
Phase 12 sidecar	**PARTIALLY ADMITTED**	75/102 tasks under the stricter all-call suffix criterion.
Phase 12B sidecar	**ADMITTED**	102/102 tasks under the repair-aware governance ruling.
Ontogenia risk	Open, non-blocking	Monitored model-output quality risk.

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